One completed AMC23 case: LGSD preserves the crucial +x

Qwen3-8B · episode 64 · same query and decoding settings · neither generation hit the token cap

PROBLEM

Find the least-degree rational polynomial P satisfying $P \equiv x + 2 \pmod{x^2 + x + 1}$ and $P \equiv 2x + 1 \pmod{x^2 + 1}$.

Question: what is the sum of the squared coefficients of P?

LGSD

CORRECT

Compact derivation

$P = (x^2 + x + 1)(mx + n) + x + 2$

$= mx^3 + (m + n)x^2 + (m + n + 1)x + (n + 2)$

$P = (x^2 + 1)(px + q) + 2x + 1$

Matching coefficients gives $n = 1,\ m = 1,\ q = 2.$

Therefore $P = x^3 + 2x^2 + 3x + 3.$

$$1^2 + 2^2 + 3^2 + 3^2 = \mathbf{23}$$

OPSD

WRONG

Where the saved response goes wrong

$P = (x^2 + x + 1)(mx + n) + x + 2$

OPSD writes the x coefficient as $m + n$**.**

It should be $m + n + 1$: the remainder contributes $+x$.

That slip gives $P = x^3 + 3x^2 + 3x + 4.$

$$1^2 + 3^2 + 3^2 + 4^2 = \mathbf{35}$$

LGSD 3,585 generated tokens · completed naturally

OPSD 5,156 generated tokens · completed naturally

Saved outputs: AMC23 · one sample per query · temperature 0.6 · top-p 0.95 · top-k 20 · seed 54,486 · max 10,240 tokens.

Derivations are condensed for readability; parsed answers, token counts, correctness, and non-truncation status are exact.

LGSD is logged as TRSD